PARKINSON'S PROGRESSION MARKERS INITIATIVE

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# Evaluation of TMPRSS5 (Spinesin) in the CSF as a potential biomarker for early Parkinson's disease diagnosis and progression

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## Summary

Based on a targeted proteomics screen we found that Transmembrane protease serine 5 (TMPRSS5, aka Spinesin) was upregulated in the cerebrospinal fluid (CSF) of idiopathic Parkinson´s disease (iPD) patients compared to healthy controls (HC). We have confirmed these findings in two independent cohorts using an in-house developed and validated immunoassay.
By analyzing TMPRSS5 levels in this cohort, we seek to extent the CSF dataset for iPD vs. HC as well as to generate first data on early iPD and SWEDD patients to evaluate the potential of TMPRSS5 as a diagnostic biomarker for early iPD. In addition, we plan to investigate if the marker can reflect disease progression and changes intra-individually during the 3-year period as well as whether levels of the marker at baseline can predict disease progression.
For this purpose we will use an in-house developed ELISA-like immunoassay on the ELLA platform (ProteinSimple) based on the customizable open cartridge format, recombinant human TMPRSS5 as calibrator and two commercially available anti-human TMPRSS5 antibodies. In this assay system a digoxigenized capture antibody gets trapped via immobilized anti-digoxigenin antibodies. After binding to the capture antibody, the analyte is detected by a biotinylated detection antibody that is subsequently visualized via fluorescently labeled streptavidin.

## Method

The 48-Digoxigenin Cartridge Kit (ProteinSimple, Cat.No. 952927) was used according to the manufacturer´s instructions. In brief, 1 mg/mL of the monoclonal anti-human TMPRSS5 capture antibody (R&D Systems, Cat.No. MAB24951) was digoxinized in PBS with 75 mg/mL sodium bicarbonate (Sigma-Aldrich, Cat.No. S8875) and 0.67 mg/mL digoxigenin-NHS ester (Sigma-Aldrich, Cat.No. 55865) for 1 hour at room temperature. In addition, 1 mg/mL of the polyclonal anti-human TMPRSS5 detection antibody (Invitrogen, Cat.No. PA546992) was biotinylated in PBS with 75 mg/mL sodium bicarbonate and 1 mg/mL Biotin-XX (Sigma-Aldrich, Cat.No. B3295) for 1 hour at room temperature. Subsequently, free labeling reagent was removed by means of Zeba Spin Desalting Columns (Thermo Fisher, Cat.No. 87766). Antibody concentrations were determined via absorbance at 280nm using a Nanodrop One spectrophotometer (Thermo Fisher).

The 48-well Digoxigenin Cartridge was loaded according to the manufacturer´s protocol and analyzed using the ELLA automated immunoassay system (ProteinSimple). Briefly, digoxigenin-conjugated capture antibody (3.5 µg/mL) was added in 50 µL RD buffer (ProteinSimple, Cat.No. 895182) into the respective "C"-wells and biotin-conjugated capture antibody (2.5 µg/mL) was added in 50 µL RD buffer into the respective "D"-wells. A total volume of 50 µL was loaded per sample. CSF samples were diluted 1:40 in SD 10 buffer. All samples were randomized over 5 cartridges and each cartridge included a seven-point, 3.16-fold calibration curve of recombinant human TMPRSS5 (R&D Systems, Cat.No. 2495-SE-010) ranging from 1000 – 0.001 pg/mL prepared in SD10 buffer (ProteinSimple, Cat.No. 896097), as well as 4 blanks, 2 PPMI CSF pools and 6 internal AbbVie controls. The AbbVie controls consisted of 3 different dilutions of two independent CSF pools.

The absolute TMPRSS5 concentration for each sample was interpolated from the RFU values based on the calibration curve after applying a logarithmic, five-parameter sigmoidal fitting model. The ELLA assay generates 3 technical replicates out of one sample, as it is analyzed in 3 separate capillaries. No sample had to be re-measured, as none exhibited a CV greater than 30% between the technical replicates, nor a concentration outside of the dynamic range of the assay. The maximum variability between a control pool measured on each plate was 18.2%. The experimenter was blinded regarding the cohort samples.

## About the Authors

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